EXPERIMENT 5

Demonstrate transformation, scaling, binning, fixing skewed values and sampling.

AIM

To demonstrate data preprocessing techniques including feature engineering, Min-Max scaling, income binning, log transformation for skewed data, and random sampling on the California Housing dataset.

PROCEDURE

1. Import pandas, numpy, matplotlib, MinMaxScaler, and fetch_california_housing.
2. Load the California Housing dataset.
3. Create a DataFrame with feature names.
4. Take a sample of 1000 rows using sample().
5. Create a new feature 'rooms_per_household' = AveRooms / HouseAge.
6. Apply MinMaxScaler to normalize all features.
7. Bin 'MedInc' into 5 income categories using pd.cut().
8. Plot histogram of Population column to observe skewness.
9. Apply log transformation using np.log1p().
10. Plot histogram after log transformation.
11. Draw a random sample of 5% of the full dataset.
12. Print the shape of the random sample.

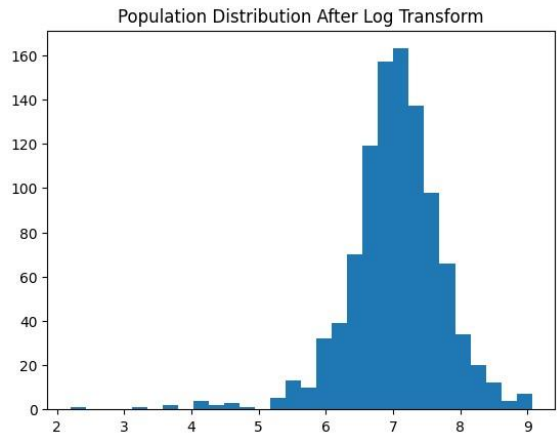
CODE

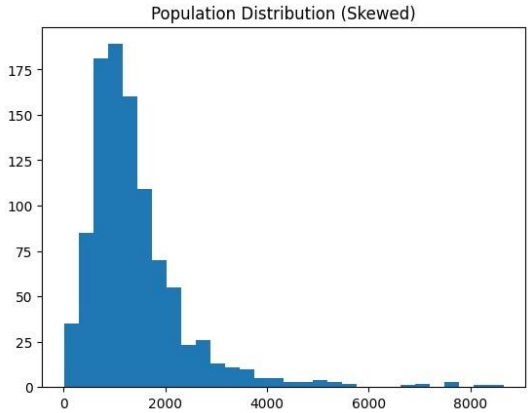
```
import pandas as pd import numpy as np import
matplotlib.pyplot as plt from sklearn.preprocessing
import MinMaxScaler from sklearn.datasets import
fetch_california_housing

housing = fetch_california_housing() data =
pd.DataFrame(housing.data, columns=housing.feature_names)

sample_data = data.sample(n=1000, random_state=42)

# Feature Engineering
sample_data['rooms_per_household'] = (
sample_data['AveRooms'] / sample_data['HouseAge']
)

# Scaling
scaler = MinMaxScaler() scaled_data =
pd.DataFrame(
scaler.fit_transform(sample_data),
columns=sample_data.columns
)

# Binning
sample_data['income_bin'] = pd.cut(
sample_data['MedInc'],     bins=5,
    labels=['Very Low','Low','Medium','High','Very High']
)
```

```
# Log Transformation (fixing skew)
plt.hist(sample_data['Population'], bins=30)
plt.title('Population Distribution (Skewed)') plt.show()

sample_data['log_population'] = np.log1p(sample_data['Population'])
plt.hist(sample_data['log_population'], bins=30)
plt.title('Population Distribution After Log Transform') plt.show()

# Random Sampling
random_sample = data.sample(frac=0.05, random_state=1) print(random_sample.shape)
```

OUTPUT

```
Sample Shape: (1000, 8)
After Feature Engineering, Columns include: rooms_per_household Income
Bins (head):
  MedInc 1.6812 → Very Low
  MedInc 2.5313 → Very Low

Random Sample Shape: (1032, 8)
```





RESULT

Feature engineering, scaling, binning, and log transformation were successfully applied. The skewed population distribution was normalized using log1p transformation. MinMax scaling brought all features to [0, 1] range and random sampling extracted a 5% stratified sample.